How Much Resolution is Enough?

Digital cameras have many settings that specify the resolution at which it captures images. The questions are: which is the right resolution and is there a single best resolution? The answers depend on where you want to eventually use your images. Are you going to simply view them on a computer screen or do you want to take prints?

Viewing on monitors: Suppose you have a 2.1 megapixel camera which offers a maximum image resolution of 1600x1200 at 72 dpi. This means that on your monitor, you will see a full screen image if you are running your monitor at a resolution of 1600x1200, because monitors have a resolution of 72 dpi. The size of this image is obtained by scaling the dimensions of the image to the resolutions. Therefore, at 72 dpi, you would have a 22.2x16.6 inch image. This is how large your image would appear on your Web page—quite huge.

Image of a subject created at 72 dpi. Notice the pixellation in the text

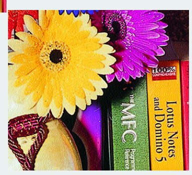

Image of a subject created at 300 dpi. The text is sharper

Print applications: The magic resolution at which most printing happens is 300 dpi. This is the minimum you should print at for acceptable results (see the quality difference in the image)—the higher the resolution you print at, the better the quality. Therefore, for a 4x6 inch print at 300 dpi, you would need to shoot images at a resolution of 1200x1800 (multiplying the size in inches by 300). Taking the same camera as before, if you were to shoot at the highest resolution of 1600x1200, you would obtain a 5.3x4 inch print image at 300 dpi (dividing the resolution by 300). This tells you that you will definitely not be able to produce 300 dpi A4-sized (8.26x11.69 inch) prints with a 2.1 megapixel camera as it cannot capture images with a high enough resolution.

Kodak DX3600, which had a rather unusual 1800x1200 pixel resolution. This was due to its 2.3 megapixel CCD as compared to the other cameras in this category which had CCDs ranging between 2.11 and 2.14 megapixels. With these image resolutions, it is evident that 2+ megapixel cameras are suited to applications where you need images not bigger than post card size since they can only deliver 5.3x4 inch prints at a 300 dpi resolution.

The optical system: The optical system is one of the most critical parts of a digital camera—to a large degree it dictates the quality of the images captured. This is especially true when applied to the zoom function. There are two types of zoom methods used in digital cameras, optical and digital (optical zoom delivers better performance). Implementing the optical zoom feature in a camera involves precision mechanics and lens elements that subsequently increase the cost of the camera. Therefore, having an optical zoom in a camera that is reasonably priced is a very big advantage. On the other hand, the simplest and cheapest lens to implement in a camera is a 'fixed-focus' lens with no zoom function, which offers average performance in all types of focusing situations.

The two cameras that stood out in the sub-2.0 megapixel range were the Kodak EasyShare DX3215 and the Olympus Camedia C-1 with optical zooms of 2x and 3x respectively. The Casio GV-10 and the Fuji FinePix A101 featured only 2x digital zoom functionality. The D-Link DSC-350 and the Benq DC300 did not have any zoom capability as they were based on fixed lenses.

The camera that shone in the 2+ megapixel category was the Olympus C-2100 Zoom. Offering a 10x optical zoom and an amazing 27x digital zoom, this camera can really let you get up close and personal to your subject! With such a powerful zoom, the image would tend to be shaky if no tripod is used. To counter this, the camera has an image stability feature that helps compensate for blurriness when used without a tripod.

Another camera that stood out here was the Casio QV-2800UX. It has an 8x optical zoom and 4x digital zoom and was built around a small form factor, just like the Fuji FinePix 2800 Zoom, which sports a 6x optical zoom and a 2.5x digital zoom. While not all the cameras in this category featured optical zoom capabilities, they all featured digital zoom with factors ranging from 2.5x to 5x.

Shutter speed: The capability to control the shutter speed of the camera lets you exercise a range of creative options. You can shoot sporting events or undertake

Kodak EasyShare DX3215

Based on a 1.3 megapixel CCD, this camera embodies simplicity. It is for those who do not want to be concerned with the complexities of photography. It sports a chunky but comfortable form factor, though the protruding lens does make it rather cumbersome when kept in your shirt pocket. The Kodak Picture Software simplifies the process of working with digital images. Another very distinguishing feature is the highly graphical user interface—it was one of the most colourful and easy to understand. A switch at the back of the camera changes its mode from image playback to video recording and image capture. The camera has a video-out port located next to the USB interface. It features a 2x optical and 2x digital zoom and 8 MB of internal memory with a Compact Flash expansion slot. To transfer photos, you can connect the camera to the computer either through the USB interface or through a docking station that also serves to recharge the bundled Ni-MH batteries.

With exceedingly simple usability and very commendable performance, the Kodak DX3215 is a good choice for a first foray into the world of digital photography.

Price: Rs 16,800

Kodak EasyShare DX3215	B
Features	
Ergonomy	
Performance	
Warranty & support	
Value for money	
OVERALL	

\+ Very simple controls, integrated memory with expansion slot
– Chunky design